

RSM — The Structural Chain (v9.3)

The implicit register: the structural requirement of logic, stated without measurement, units, or chart. What instantiates it — physical, biological, textual — belongs to the explicit register and is not this chain's subject. Not sealed.

0. The two registers

Every claim in this project belongs to one of two registers, and the division is decidable before any result arrives.

- **Implicit** — what follows from P1, P2 and the frame Definition with no measurement, no units, and no drawing. The variables below are of this kind. X and Y are not names; they are the slots two conjugate modes occupy.
- **Explicit** — anything requiring a rendering to state: coordinates, a metric, a sign convention, a quantity, an instance. Physics, biology, and any named pair (有/無, up/down) live here.

The implicit register fixes the structure; it fixes no parameters. Two systems can instantiate one structure and differ in every number — a branching law with exponent two and one with exponent three are the same logic under different parameters, and the implicit register is what they share. It follows that no quantity is ever owed by this chain, and no quantity ever confirms it.

This division is not a defence built after the fact. A claim is implicit only if it can be derived without a chart; anything else is explicit whether it succeeds or fails.

Premises

P1 — the Conditional. Reality is infinitely vast and infinitely divisible: no terminal resolution in either direction — no edge outward, no floor inward.

P2 — the Bridge. What obtains is distinguishable. Contrapositively: what is not distinguishable does not obtain.

Definition (frame). A frame R_n is the coinherence of one distinction: two conjugate modes, their reciprocal relation, their balance, and their origin — arriving together or not at all. One constitutive pair per frame.

Corollary. Every distinction constitutes a frame; distinction beneath a frame's unit is therefore another frame, not a smaller item within this one.

Everything below is what P1 and P2 jointly require. Where a further assumption operates, it is named in the open items.

1. The total paradox: P_0

P_0 is the master conjugate pair at equality: **distinction and indistinction**, each constituted by the other, standing equal and cancelling. The same condition says itself three ways — distinction equals indistinction; obtainedness equals non-obtainedness; coherence equals incoherence. It is not indistinction: indistinction is one side of the pair, and P_0 privileges neither.

It cannot obtain, because at the total cancellation nothing is distinguishable and obtaining would not differ from not obtaining. It cannot be asserted without contradiction, since stating the equality distinguishes its terms. And it is not a state that fails to obtain but a condition that fails to cohere.

Its presenting faces are the **empty face** (collapse via emptiness) and the **full face** (collapse via fullness). Neither is a number. In native coordinates the empty face sits at nothing and the full face at the unbounded end — never at any frame's unit. The two faces are the two counterfactual terminals of one journey along the gradient: toward all-form-no-void and toward all-void-no-form. Neither is reached; the journey is one. Both are excluded in-frame, at the gradient's two ends, by the law's own shape (§4).

P_n is the same species at frame scope: a frame's own constitutive pair at equality.

Indistinction does not obtain in a frame built on distinction. Where up equals down there is no height. Where before equals after there is no time. Where void equals form there is no reality. In each case the frame *is* the distinction, so equality removes what the frame was — and note the strength of that: no height, not height zero. Height zero is a value inside a frame that has height.

The difference between the two scopes is what survives the collapse. At P_n the frame has no locus there, but the frame obtains elsewhere and other frames obtain. At P_0 there is no elsewhere, because the pair that fails is the one every pair presupposes.

2. The ground

P_0 cannot obtain. Nor can indistinction on its own: it is one side of the master pair, and by P2 what is not distinguishable does not obtain. Both are excluded, so what obtains is distinguishable — at least one distinction, an origin, a frame: O_1 , R_1 . The step runs through P2 alone. P1 is the Conditional and asserts no existence. The ground relation is the

recursion relation at zero scope: every origin is a paradox read from below, and O_1 reads P_0 . Because P_0 is unframed, R_1 has no parent: each obtained tree is rooted. The ground's mechanism (identity through opposition — required directly by the impossibility) differs from every later frame's (generation at a paradox); both are real, and the structure needs both.

At least one is what the impossibility gives. Nothing here counts the roots, and nothing here excludes more than one.

The two senses of smaller. Two distinct relations must not share a word:

- **Finess** — motion along the gradient toward a mode's small end. In-frame, unbounded, terminal excluded by the law.
- **Descent** — the constitution of a child. Across-frame, inward only, not motion along anything.

Vastness and fineness are the gradient's two ends, both in-frame, exchanged by the mode-swap. Descent is neither of them.

3. The coinherence

- A distinction entails its two co-obtaining modes (X , Y) — one distinction, two modes, each constituted by the other; nothing prior was divided.
- **The modes are magnitudes. The conjugate of a mode is another mode, not its negative.** A negative would name something that both obtains and anti-obtains; what holds instead is that a mode is incoherent without its contrasting mode. A sign is an **indexed comparison** — this, compared to what it is being compared to — and its implication varies with the reference chosen. Sign therefore belongs to the explicit register throughout.
- A mode can be noticed without its conjugate having been measured; its **amount** exists only as a ratio to its conjugate. Existence-dependence is not measurement-dependence.
- Both modes are measured from one reference — the point at which both would vanish. That is the origin O : a locus, never a position; nothing stands there.

4. The frame; the unit; the floor

1_n is the constitutive distinction's own magnitude — the conserved product of *the* pair that constitutes frame n — and simultaneously the frame's minimal unit: nothing beneath the constitutive magnitude obtains *within* frame n , because any distinction beneath it is, by the Corollary, another frame. "Sub- 1_n " is unit-relative typing, not size: magnitude comparison

across frames is not defined, and the child is not smaller — the child is the approach that does not finish, continuing. The floor is the boundary of the frame's own definition, not an added restriction.

The law. The two modes' amounts are locked: their product is conserved, $X \cdot Y = 1_n$. A conserved sum fails twice — it admits the terminals (one mode at nothing, the other at everything: P_0 's faces as attainable values) and it caps both modes, contradicting infinite reach. The product's curve has no endpoints and no ceilings; the exclusion of the impossible is the law's own shape, not a rule imposed on it. The constant is the unit, and the unit is the frame's reality: things differ in their proportion of the modes; reality is the invariant the proportions hold.

Scale. Within a frame, scale is motion along the curve: rescaling one mode by λ and the other by $1/\lambda$, product invariant. No rescaling reaches the center. Scale is within-frame; the lineage is not scale.

5. The two centers, and the line between them

O — the co-vanishing locus. Where both modes would reach operational absence together. Nothing on the curve approaches it; the curve bends around it.

B_n — the balance line. The whole locus $X_n = Y_n$. It reaches from O_n outward and passes through the gradient's paradoxical center. **No point of B obtains.** At any point of B the two modes stand in no proportion, so nothing distinguishes them; by P_2 the condition does not obtain. B is a limit locus along its whole length: approached, never occupied.

P_n — the paradox. Where the gradient meets the balance line. It is the gradient's own center in the sense the origin is not: O is the center the curve bends around and never reaches, P_n is the center the curve has — its one point fixed by the mode-swap. P_n does not obtain, for the reason B does not. What frame n has instead is a resolution limit: **at frame n's resolution the two amounts are not distinguishable near P_n .** That is a fact about the frame's discrimination, not about a locus being reached. Beneath the floor the difference is there and the frame cannot state it.

The disconnection. Since no point of B obtains, removing B leaves the X-dominant and the Y-dominant regions of the frame with no path between them. Within frame n the two mode-dominant regions are *related* — each is the other under the mode-swap — but not *connected*. No traversal joins them.

6. The engine

Continuity entails the balance line, and the gradient meets it. Denial A (origin): the locus cannot be frame n's origin — one constitutive pair per frame, and n has no spare pair to

grant it. Denial B (exactness): the condition cannot obtain at all, since at exact equality the frame is not there. Simultaneously, locally.

The demand is satisfied in the one mode of existence left to such a center: **existence-as-organizing-locus** — in-frame, as the locus circulation organizes around; trans-frame, as the origin a new structure organizes around. **The child frame is the pursuit of exactness continuing below the floor:** the residue of imbalance beneath 1_n — the not-quite-balance the frame cannot state — is sub-unit distinction, hence a frame, whose origin is this very locus read from below. Recursion is exactness, permanently deferred.

Generation has no address. Constitution is not sited and not timed. The residue of imbalance is standing, so the child is entailed, not triggered. Nothing occurs *at* exact balance, because there is no there.

Rotation. The frame's two mode-dominant regions are related without being connected (§5), and the relation cannot be carried by a path. It is carried by revolution: **G_n revolving about B_n .** A half-turn about B exchanges the modes — the mode-swap performed as motion rather than as relabeling — so revolution relates the two regions without anything passing through the locus that cannot be occupied. "No through, only around," with the going-around structurally forced rather than merely preferred. The plane's own exchange is reflection across B, which reverses orientation and is no motion; realizing the swap as orientation-preserving continuous motion is what requires a direction the plane does not have. Orientation is linear-structure content, not measure content, so the requirement is stated without a metric.

Rendering guard. A drawing shows a smooth curve through an unproblematic point; the implicit facts (no origin here; nothing obtains here) have no rendering. The drawing's verdict never overwrites the logic's.

7. Recursion

One locus, two frame-readings: a paradox in the parent's register, an origin in the child's — **$O_{(n+1)}$ reads P_n .** And one line, two frame-readings: **$yAxis_{(n+1)}$ reads B_n .** The identification is of type, not merely of position: an axis is a locus where a mode is exactly zero and which therefore does not obtain; B is a locus where the modes are exactly equal and which therefore does not obtain. Same species of object. What the parent cannot resolve is what the child measures from.

Orientation. B is not selected by the parent — P_n lies on B by construction, so a line through the child's origin is constituted, not chosen. What remains free is everything about that line: the rotational freedom about B is the **ring** of orientations, no one privileged. What obtains is always one selection (all at once would be saturation), and different selections are different branches. Every obtained frame therefore has at least one orientation, and it is the

inherited one.

The nesting is a tree, not a ladder — and a rooted tree: index-relationality holds within it (every frame reads itself as “n”; the labels are gauge), but no relabeling makes a root a child, because a root’s parent-side is P_0 , an impossibility, not a frame.

What a child shares with its parent. Not magnitude (comparison across frames is undefined), not orientation about B (nothing selects it), not content, not substance. What is shared is the frame-form — every frame is a frame — and, between this child and this parent, the paradox itself: the parent’s inexactness is the child’s reference. The relation is constitutive and standing, not a transfer and not an event.

Words that assert what the framework denies

A standing list, not a procedure. Agentive verbs for structural facts. Transfer-language for non-events. Removal narratives for constitutive exclusions. Staging conjunctions for simultaneous constitution. Reified infinities. Occupancy language for the loci. Address-language for constitution. “Negative” for a conjugate. Any noun that turns a paradox into a place.

The involution. Two are available: the mode-swap ($X \leftrightarrow Y$, fixed set B) and the sign-flip (componentwise negation, fixed point O). The mode-swap is structural; the sign-flip is explicit. A negative would name something that both obtains and anti-obtains, whereas a mode is incoherent without its contrasting mode. And a half-turn about B sends $X \rightarrow Y$ and $Y \rightarrow X$ — the mode-swap is a rotation, a motion the structure admits and requires; componentwise negation reverses orientation and is no rotation at all.

Open items

1. **The open closures** — the reciprocity requirement’s native-magnitudes step; anchoring-by-constitution; the ground’s self-sufficiency candidate.
2. **The cross-frame unit relation** $1_n \leftrightarrow 1_{(n+1)}$ — whether any quantitative relation accompanies the constitutive share of §7. An explicit-register question.
3. **Dimension count** — revolution about B requires a direction the plane lacks (§6). How many such directions the structure requires is not settled.
4. **Branching beyond the balance.** Generation has no address, so the question is whether any relation other than the constitutive pair can be inexact in the way that constitutes a child. Candidate, unruléd: it cannot — the frame has one constitutive pair, any conjugate equality within it reduces to the same balance, and the other points of B belong to other frames.

v9.3 — the chains state the framework as currently known; the editorial record holds how it got here.